



Embedding electronic perpetual motion into single-atom catalysts for persistent Fenton-like reactions

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Edited by Alexis Bell, University of California, Berkeley, CA; received September 16, 2023; accepted November 30, 2023

In our quest to leverage the capabilities of the emerging single-atom catalysts (SACs) for wastewater purification, we confronted fundamental challenges related to electron scarcity and instability. Through meticulous theoretical calculations, we identified optimal placements for nitrogen vacancies (Nv) and iron (Fe) single-atom sites, uncovering a dual-site approach that significantly amplified visible-light absorption and charge transfer dynamics. Informed by these computational insights, we cleverly integrated Nv into the catalyst design to boost electron density around iron atoms, yielding a potent and flexible photoactivator for benign peracetic acid. This exceptional catalyst exhibited remarkable stability and effectively degraded various organic contaminants over 20 cycles with self-cleaning properties. Specifically, the Nv sites captured electrons, enabling their swift transfer to adjacent Fe sites under visible light irradiation. This mechanism accelerated the reduction of the formed “peracetic acid-catalyst” intermediate. Theoretical calculations were used to elucidate the synergistic interplay of dual mechanisms, illuminating increased adsorption and activation of reactive molecules. Furthermore, electron reduction pathways on the conduction band were elaborately explored, unveiling the production of reactive species that enhanced photocatalytic processes. A six-flux model and associated parameters were also applied to precisely optimize the photocatalytic process, providing invaluable insights for future photocatalyst design. Overall, this study offers a molecule-level insight into the rational design of robust SACs in a photo-Fenton-like system, with promising implications for wastewater treatment and other high-value applications.

computational predictions | electronic perpetual motion |
nitrogen-vacancy-rich iron single-atom | photo-Fenton-like reaction |
reaction mechanism

Fenton-like reactions, driven by peracetic acid (PAA), present a compelling approach to organic wastewater treatment, characterized by several unique advantages. These include the avoidance of toxic by-products often found in persulfate-activated systems (1–3), a lower O–O bond binding energy (159 kJ mol^{-1}) than H_2O_2 [213 kJ mol^{-1} (4–6)], and the potential for residual oxidants and products to serve as a carbon source for microorganisms in subsequent biological treatment (7–13). Moreover, PAA can readily generate active and persistent organic radicals such as $\text{CH}_3\text{CO}^\bullet$, $\text{CH}_3\text{COO}^\bullet$, and $\text{CH}_3\text{COOO}^\bullet$, which have longer half-lives and are highly selective in degrading organic pollutants (14, 15). While current PAA-activated systems primarily rely on transition metal ions [e.g., Fe^{2+} (7), Co^{2+} (16)] and UV-assistance (11), these approaches often pose challenges related to secondary pollution and energy demands (17, 18). Due to its affordability and wide usage in conventional Fenton systems (19, 20), iron offers a promising alternative. Specifically, a visible-light-active Fe-based system could be established without additional energy input (21), emphasizing the selection of suitable substrates.

Importantly, these substrates must exhibit strong stabilization against iron ions, and N-rich materials can provide this through robust Fe–N coordination, preventing metal precipitation (22, 23). Drawing upon previous studies (24, 25), carbon nitride ($\text{g-C}_3\text{N}_4$), with its abundance of N atoms and associated lone pairs, offers an attractive solution to stabilize metal sites, along with benefits such as visible-light absorption, easy synthesis, low-cost, and stability (26, 27). Nonetheless, $\text{g-C}_3\text{N}_4$ suffers from drawbacks like slow charge separation, susceptibility to e^- – h^+ pair recombination, and limited light utilization (28, 29)—issues that integration with Fe^{3+} may help overcome (30, 31). Traditional Fe doping may limit iron utilization and catalytic activity, calling for a more nuanced approach.

The advent of single-atom catalysts (SACs) has opened frontiers in catalysis, offering the promise of maximum metal utilization, tunable coordination, and robust metal-support interactions (32–34). In the case of iron (Fe) SACs, the optimal exposure of atomic Fe sites can catalyze reactions more efficiently, increasing oxidation capacity. However, challenges such as insufficient electronic sources and Fe sites instability must be addressed.

Significance

The recent development of single-atom catalysts (SACs) as oxidant activators represents a significant advancement in heterogeneous Fenton-like systems. However, the progression toward creating robust and efficient SACs is marked by numerous challenges. Herein, our research delineated a strategy, employing forward theoretical prediction to intricately tailor nitrogen vacancies and single-atom iron sites within $\text{g-C}_3\text{N}_4$ -based catalysts. By utilizing visible light as a switch for continuous electron input, our efforts culminated in a powerful system as an effective photoactivator of peracetic acid using well-regulated SACs, uniquely activating molecular oxygen. This impressive system featured a photoswitching self-cleaning and demonstrated remarkable stability. Our groundbreaking work establishes a theoretical foundation for crafting high-performance SACs for wastewater purification and other high-value reactions.

Author contributions: F.C. designed research; F.C. and Y.-J.S. performed research; F.C. contributed new reagents/analytic tools; F.C., Y.-J.S., X.-T.H., C.-W.B., Z.-Q.Z., P.-J.D., X.-J.C., and Q.Y. analyzed data; and F.C., Y.-J.S., and H.-Q.Y. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2314396121/-/DCSupplemental>.

Published January 18, 2024.

Additionally, while carbon-based metal- N_x coordination structures show high reactivity, they may suffer from reduced activity over time due to the loss or agglomeration of single-atom metal sites. Nitrogen vacancies (Nv) serve as electron trap centers and can be formed by adequately opening the heptazine ring of $g-C_3N_4$ (35). Integrating Nv as a constant source of electrons for Fe sites can significantly enhance stability and concurrently lead to surface defects and uneven electron distribution. The accumulation of electrons in the Nv region means that fewer electrons can directly contribute to the metal sites, allowing for the utilization of a visible-light switch as an additional energy source. This innovation may further contribute to the stability of metal sites.

Given that $g-C_3N_4$ possesses six extended electron pairs of nitrogen sites capable of immobilizing Fe, the simultaneous introduction of Fe single-atom and Nv sites onto $g-C_3N_4$ holds potential as an efficient PAA photoactivator. Should Nv be strategically positioned near the Fe single-atom site, photogenerated electrons may be captured initially by the Nv sites. This process could then facilitate a direct transfer and accumulation of electrons at adjacent Fe sites, assisted by visible light, thereby enhancing PAA activation through continuous electron injection. However, crafting the optimal synergy between these dual sites represents a significant challenge. To address this, theoretical calculations could be instrumental in predicting the best locations for the Nv and Fe sites, refining the catalyst's design. Such predictions can include optical properties and charge transfers for various site interventions. Furthermore, the underlying relationship between the dual sites, catalytic activity, and stability still requires more detailed investigation and understanding.

Consequently, the first objective of this work was to perform multiscale simulations to guide the design of Nv-rich Fe SACs. Subsequently, bisphenol A (BPA), a common endocrine disruptor, was chosen as the target to assess the photocatalytic activities, both in the presence and absence of Nv and Fe sites. A thorough examination of active species development was performed using quenching experiments and electron spin resonance (ESR) measurements, while theoretical calculations and mass spectrometry analyses were utilized to clarify the underlying catalytic mechanisms and pathways of contaminants. The findings of this work demonstrate the potential of visible-light-activated PAA in conjunction with predesigned, well-engineered Fe SACs. The innovative concept of electron input introduced here offers a promising avenue for the cost-effective, long-lasting degradation of organic pollutants in SACs-based AOPs.

Results and Discussion

Design of Nv-Rich Fe SACs Guided by Theoretical Calculations.

The computational simulations were first conducted to predict the most suitable Nv position. To determine the optimal Nv sites on FeCN-bulk coupled to Fe- N_4 , we selected and optimized six Nv site models with different relative distances from the Fe site (Fig. 1 A–D and *SI Appendix*, Fig. S1 and Table S1). The results based on the optimized models show that the lowest formation energy ($E_f = 0.00$ eV) could be achieved when the isolated Fe atom was coordinated to 4 N and bound to the neighboring N3C-type Nv (Fig. 1B), indicating that the neighboring N3C-type Nv was the most suitable.

The migration separation efficiency of photogenerated charges in a two-dimensional conjugate plane significantly influences photocatalytic activity. Thus, work functions were then calculated to characterize the minimum energy required for electron transfer from the Fermi level to the vacuum. The surface work functions for pristine CN-bulk, CN-Nv, and FeCN-bulk-2 were

4.604 eV (Fig. 1E) and 3.968 eV (Fig. 1F), and 3.627 eV (Fig. 1G), respectively. These values suggest that combining Nv and Fe single-atom sites could reduce the energy barrier for electron escape into the vacuum, favoring the following PAA activation. Among them, when the single-atom Fe coexisted with Nv (FeCN-Nv-2, Fig. 1H), the required energy was slightly higher than FeCN-bulk-2, possibly due to electron attraction toward Nv, necessitating external energy sources like visible light, as utilized in this study.

Also, the density of states (DOS) calculations of different samples were conducted to explore the influence of Nv and Fe on the CN-bulk's electronic structure. In the case of pure CN-bulk, the valence band (VB) was dominated by nitrogen (N), and the conduction band (CB) by carbon (C) elements (Fig. 1I). Introducing Nv led to a redistribution in the electronic structure, increasing C's contribution to the VB and causing the emergence of a new intermediate state gap near the CB's bottom (Fig. 1J). This altered energy band structure, reducing the CN-bulk band gap and enhancing light absorption. Similarly, incorporating a Fe single atom resulted in its unfilled orbitals significantly contributing to the CB, promoting a shift in the band and enabling easier migration of VB electrons to the CB (Fig. 1K). Moreover, when Fe single atom site was present in CN-bulk with Nv, the unfilled empty orbitals in Fe and the C element with Nv partially contributing electrons were still the primary contributors to the CB and VB, respectively. This indicates that with the excitation of external energy, Nv electrons would primarily be transferred to the empty orbitals brought by the Fe site. These results also verified the importance of the light switch (Fig. 1L). Thus, Nv could create a new energy level and act as an electron trap, while Fe insertion bolstered oxidizing ability, favoring PAA activation.

These theoretical predictions were instrumental in optimizing the arrangement of Nv and Fe sites within the $g-C_3N_4$ framework, revealing that dual sites amplified $g-C_3N_4$'s light absorption and charge transfer abilities. Fig. 1 M–P presents a magnified view of various active sites in $g-C_3N_4$, showcasing a Fe- N_4 coordination and adjacent Nv formed in a heptazine ring near Fe sites. Then, a detailed scheme for regulating dual sites is provided. As shown in Fig. 1Q, both Nv and Fe single-atom sites were vividly exhibited. By secondary high-temperature calcination under N_2 conditions, bulk $g-C_3N_4$ could be exfoliated to form nanosheets while generating Nv simultaneously. Because a Fe- N_4 coordination could firmly anchor Fe sites, the atomically dispersed Fe sites were well incorporated into $g-C_3N_4$. Interestingly, secondary exfoliation and dual-site intervention resulted in ultrathin nanosheets of less than 4 nm, allowing more active sites (*SI Appendix*, Fig. S2). Additionally, the BET surface area, pore diameter, and volume exhibited increased values (*SI Appendix*, Table S3 and Fig. S3). These modifications facilitated photocatalytic reactions, culminating in the realization of the theoretically driven Nv-rich Fe SACs design.

Characteristics of Nv-Rich Fe SACs. Crystalline structures of different prepared catalysts were first investigated by X-ray diffraction (XRD) patterns (Fig. 2A). These analyses revealed consistent results, with prominent peaks at 2θ values of 12.84° and 27.38° , corresponding to the (100) and (002) crystal planes, respectively (23, 24, 28). Intriguingly, the (100) crystal plane was associated with an in-plane packed tri-s-triazine unit, while the (002) plane corresponded to the layer stacking of conjugated aromatic structures (36). The absence of Fe-related peaks attests to its minimal content within the structure, preliminarily indicating a possible presence in a Fe single atom (37). Complementary

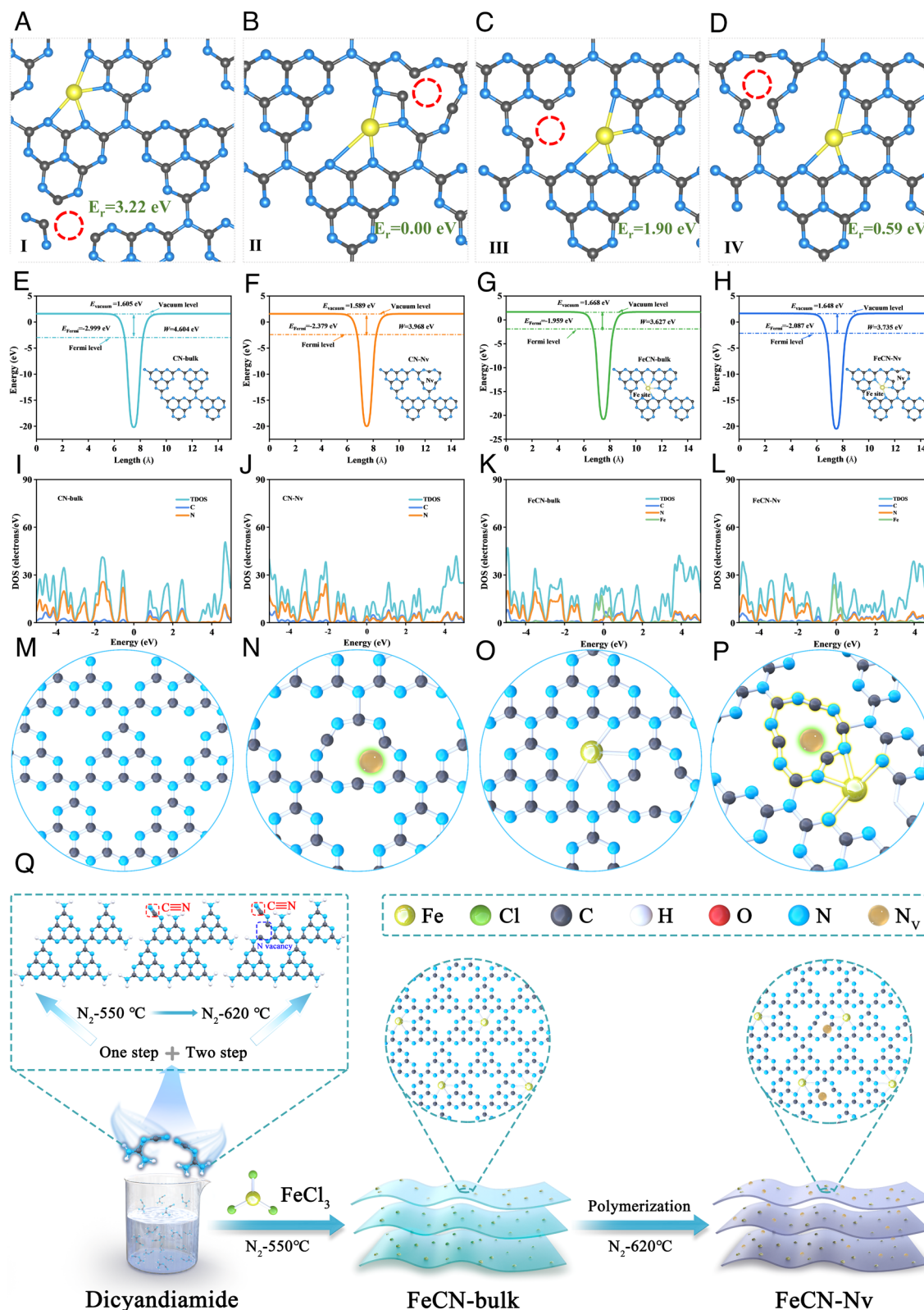


Fig. 1. Multiple predictions and synthesis of Nv-rich Fe SACs. (A–D) The relative energy at different relative positions of Fe and Nv sites. (E–H) Work functions: (E) CN-bulk, (F) CN-Nv, (G) FeCN-bulk-2, and (H) FeCN-Nv-2, respectively. (I–L) DOS: (I) CN-bulk, (J) CN-Nv, (K) FeCN-bulk-2, and (L) FeCN-Nv-2. (M–P) Schematic models of (M) CN-bulk, (N) CN-Nv, (O) FeCN-bulk-2, and (P) FeCN-Nv-2. (Q) Schematic diagram of FeCN-Nv SACs synthesis.

investigations through Fourier transform infrared spectroscopy (FTIR, Fig. 2B) yielded consistent findings with nominal variations. Specific observations included a characteristic peak at 800 cm⁻¹, indicative of the *s*-triazine units (38), and a complex

set of peaks in the region of 1,200 to 1,700 cm⁻¹, attributable to the stretching vibrational modes of aromatic C–N heterocycles (39). Peaks spanning 3,083 to 3,300 cm⁻¹ were associated with N–H or O–H stretching fluctuations, a subtlety that hints at

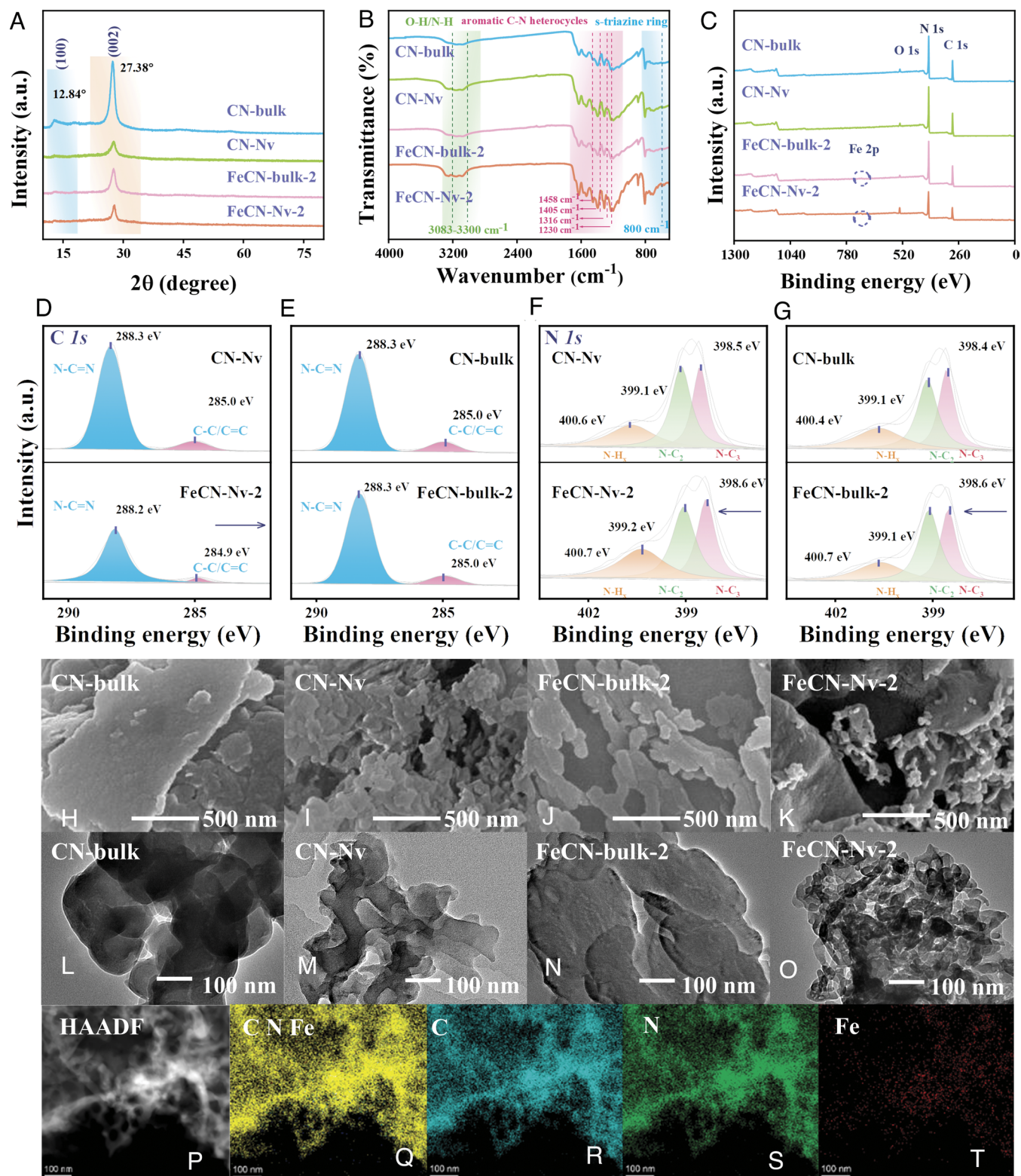


Fig. 2. Validation of basic structure and morphology of Nv-rich Fe SACs. (A and B) XRD patterns (A) and FTIR spectra of different catalysts (B). (C–G) XPS spectra comparisons: (C) survey, (D and E) C 1s, and (F and G) N 1s of different catalysts. (H–O) SEM and TEM images of (H and L) CN-bulk, (I and M) CN-Nv, (J and N) FeCN-bulk-2, and (K and O) FeCN-Nv-2. (P–T) HAADF-STEM elemental mappings of FeCN-Nv-2 with (P) original, (Q) whole, (R) C, (S) N, and (T) Fe.

partial hydrogenation of exposed N atoms within the $g\text{-C}_3\text{N}_4$ framework (40). The absence of detectable peaks and changes in Fe oxide further suggested that the robust structure of $g\text{-C}_3\text{N}_4$ and Fe sites was not embedded within Fe–O but stabilized within Fe–N. Complementing these analyses, solid ESR was employed to probe Nv (SI Appendix, Fig. S4), revealing an enhanced

Nv signal relative to CN-bulk, thus confirming the successful insertion of Nv.

X-ray photoelectron spectroscopy (XPS) was used to delineate the surface chemical compositions and states of catalysts. The survey spectra (Fig. 2C) confirmed the presence of C, N, and O elements across all samples, with Fe detected exclusively in the

targeted samples. In the C 1s spectra (Fig. 2 *D* and *E* and *SI Appendix*, Table S4), all catalysts exhibited peaks at around 285.0 and 288.3 eV, attributable to the graphitic carbon (C-C/C=C) and hybridized carbon (N-C=N) (41). The N 1s spectra (Fig. 2 *F* and *G* and *SI Appendix*, Table S5) were discernibly partitioned into three distinct peaks at 400.7, 399.2, and 398.6 eV, attributable to N-H_x, N-C₂, and N-C₃, respectively. Subsequent comparisons of N content (*SI Appendix*, Tables S6 and S7) showed an anticipated reduction when Nv was formed during secondary calcination, with the total N ratio diminishing from 54.2% (FeCN-bulk-2) to 51.9% (FeCN-Nv-2), implying the successful incorporation of Nv. The O 1s spectrum displayed a singular peak at ~532.5 eV, possibly associated with the adsorbed oxygen during the test (*SI Appendix*, Fig. S5 and Table S4).

Supplementing these above analyses, scanning electron microscopy (SEM) and transmission electron microscopy (TEM) observations were conducted. Observations of CN-bulk (Fig. 2 *H* and *L*) in bulk form were succeeded by the production of CN-Nv nanosheets (Fig. 2 *I* and *M*) through secondary calcination. The strategic introduction of Fe was found to reduce the CN-bulk thickness (Fig. 2 *J* and *N*), and additional N₂ calcination yielded even thinner, Nv-rich Fe SACs (Fig. 2 *K* and *O* and *SI Appendix*, Figs. S6 and S7), thereby amplifying the available reaction area. An aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) further disclosed the atomic-level dispersion of Fe sites on CN-Nv. A proliferation of bright spots, suggestive of a high density of Fe single atoms, were displayed in Fig. 3 *A–D*, and marked with yellow circles. The associated EDS images reconfirmed the uniform C, N, and Fe dispersion throughout the catalyst (Fig. 2 *P–T*).

To further investigate Fe's chemical states and local coordination environment, high-resolution Fe 2p XPS spectra, synchrotron radiation-based X-ray absorption near edge structure (XANES), and extended X-ray absorption fine structure (EXAFS) were integrated. Fig. 3*E* and *SI Appendix*, Table S8 display two binding energies of Fe 2p_{3/2} and Fe 2p_{1/2} at 711.0 and 725.5 eV, respectively. Notably, the binding energy of Fe 2p in FeCN-Nv-2 was shifted to a lower level than FeCN-bulk-2, suggesting the presence of Nv. The role of Nv, in this context, might be multifaceted, acting as an electron trap that transfers electrons to neighboring Fe sites, thus increasing the surrounding electron density and favoring oxidant activation. XANES analysis provided a more nuanced view of the atomic structure of Fe species in FeCN-Nv-2. Fig. 3*F* reveals that the absorption edge of FeCN-Nv-2 was situated at a higher energy level than Fe foil and FeO, but lower than the standard spectra of Fe₂O₃ (closer). This implies that Fe existed in an intermediate valence state, oscillating between +2 and +3, with a bias toward the trivalent state. Fourier transformed EXAFS curve of FeCN-Nv-2 (Fig. 3*G*) coincided with this observation, showing a pronounced peak at approximately 1.50 Å, in harmony with the Fe-N bond found in FePc. No Fe-Fe (2.21 Å/4.51 Å) or Fe-O (1.66 Å in Fe₂O₃ and 1.44 Å in FeO) peak was found (*SI Appendix*, Table S9), thus negating the presence of Fe clusters or related oxides.

The results presented herein prove that Fe sites were intricately embedded in CN-Nv in an atomic-level Fe-N configuration. Detailed analysis of the fitted curves (Fig. 3*H*) and the EXAFS fitting parameters at the Fe K-edge (*SI Appendix*, Table S10) revealed a coordination number of 4.2 for the Fe-N bond in the FeCN-Nv-2. Moreover, wavelet transforms (WT) (Fig. 3 *I–L*) further supported this finding, corroborating the methodical design of atomic-level pyrrolic N-rich Fe-N₄ sites.

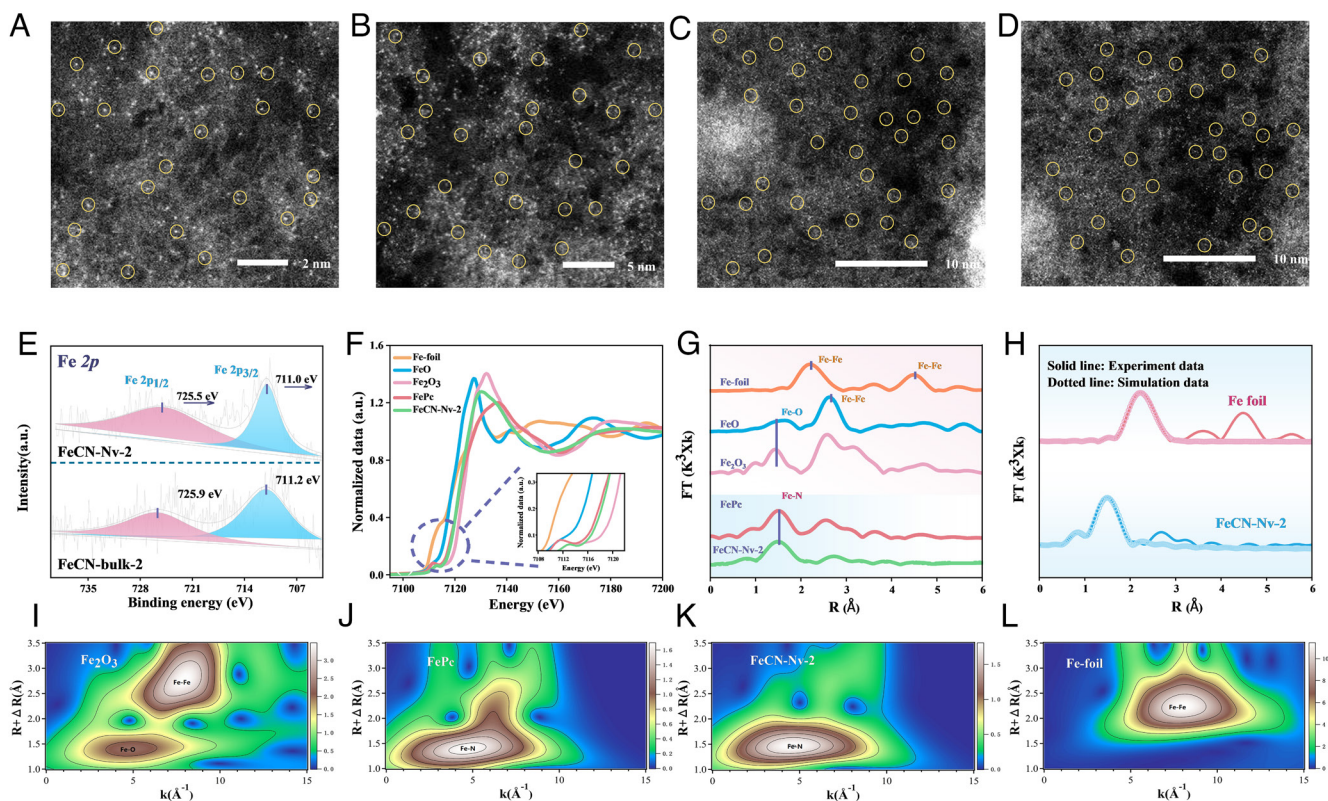


Fig. 3. HADDF and XAFS characterization of Nv-rich Fe SACs. (*A–D*) HAADF-STEM images at different magnifications of FeCN-Nv-2 (the distributed bright dots represent Fe SA). (*E*) Comparison of the high-resolution Fe 2p XPS spectra of FeCN-bulk-2 and FeCN-Nv-2. (*F–H*) Normalized Fe K-edge XANES (*F*), K³-weighted Fourier transform EXAFS (*G*) of Fe foil, FeO, Fe₂O₃, FePc, and FeCN-Nv-2, and (*H*) the results of EXAFS fitting curves in R space for Fe foil and FeCN-Nv-2. (*I–L*) WT-EXAFS for K-edge for (*I*) Fe₂O₃, (*J*) FePc, (*K*) FeCN-Nv-2, and (*L*) Fe foil.

Boosted Light-Harvesting and Photoelectric Capacities of Nv-Rich Fe SACs. The optical characteristics of the synthesized photocatalysts were evaluated through UV-visible diffuse reflectance spectra (UV-vis DRS) measurements, revealing significant insights into their absorption properties (*SI Appendix, Fig. S8A*). CN-bulk displayed an absorption edge at approximately 450 nm. However, when modified with Nv sites or Fe single-atom sites, a noticeable enhancement in visible light absorption was observed, becoming even more pronounced when two sites were incorporated. Kubelk-Munk function conversion curves (*SI Appendix, Fig. S8A*) further corroborated these findings. The estimated bandgap energies (E_g) for the various photocatalysts were as follows: CN-bulk at 2.78 eV, CN-Nv at 2.73 eV, FeCN-bulk-2 at 2.60 eV, and FeCN-Nv-2 at 2.42 eV (*SI Appendix, Fig. S8B*).

SI Appendix, Fig. S8 C–F depicts the time-resolved fluorescence spectra of the materials, from which the average emission lifetimes were derived according to Eq. 1. The CN-bulk exhibited a lifetime of 7.467 ns, while the inclusion of Nv or Fe sites led to reduced lifetimes of 4.988 ns (for CN-Nv), 4.255 ns (for FeCN-bulk-2), and 3.953 ns (for FeCN-Nv-2). This reduction in lifetime points to the possibility of electron transfer from the bulk substrate to the Nv or Fe sites, opening an additional nonradiative decay channel. Consequently, Nv and Fe sites enhanced charge transfer and separation, augmenting the catalytic reactions.

$$\tau_{ave} \equiv \frac{(A_1 \times \tau_1^2 + A_2 \times \tau_2^2)}{(A_1 \times \tau_1 + A_2 \times \tau_2)}, \quad [1]$$

where A_1 and A_2 are the pre-exponential factors, and τ_1 and τ_2 indicate the long and short fluorescent lifetime.

Furthermore, the carrier transfer and separation kinetics of prepared catalysts were further corroborated by PL, CV, LSV, MS, EIS, and i - t measurements (*SI Appendix, Text S11–S13 and Figs. S9–S11*). In particular, FeCN-Nv-2 exhibited the highest current and the lowest charge transfer resistance (*SI Appendix, Fig. S11*), favoring the following activation of molecular oxygen and PAA.

Construction of a Photo-Fenton-Like System and Performance Evaluation. To ascertain whether Nv and Fe single-atom sites in g-C₃N₄ could be an efficient photoactivator of PAA, BPA was selected as the target pollutant for catalytic degradation. The results were intriguing, showing distinct differences in catalytic capacity under various conditions (Fig. 4A and *SI Appendix, Table S11*). In the absence of visible light and PAA, the sole presence of suspended particles in the designated pollutant solution resulted in a BPA removal efficiency of approximately 2.0%. This suggests that the adsorption process was of minimal significance. Moreover, employing CN-bulk with PAA solely under visible light conditions led to a modest BPA removal of 3.6%, a limitation attributable to restricted light absorption and the hastened recombination of carriers. The incorporation of either Nv or Fe sites into the CN framework notably enhanced BPA removal, achieving efficiencies of 20.9% (CN-Nv+PAA/vis) and 20.8% (FeCN-bulk-2+PAA/vis), respectively (Fig. 4B and *SI Appendix, Figs. S13 and S14*). Impressively, the system equipped with both Nv and Fe sites (FeCN-Nv-2+PAA/vis) realized a complete (100%) BPA removal under similar conditions. This efficiency significantly surpassed other systems, outperforming them by factors ranging from 9.6 (FeCN-bulk-2+PAA/vis) to 55.6 (CN-bulk+PAA/vis) times. Importantly, even in the absence of light, the FeCN-Nv-2+PAA/dark system achieved an 80.1% BPA removal. Nonetheless, this rate was substantially lower compared to the visible light-assisted

system, reinforcing the critical role of a visible-light switch (*SI Appendix, Fig. S15*). Consequently, the synergistic integration of Fe and Nv sites within CN framework demonstrated remarkable efficacy in rapidly activating PAA under visible light conditions. Leveraging the electron-rich characteristics inherent to Nv sites, this innovative approach introduced promising possibilities for engineering other transition-metal-based SACs (Fig. 4C). The outstanding performance of FeCN-Nv-2+PAA demonstrated its extensive applicability across various applications.

In an initial examination, we concentrated on assessing the influence of catalyst dosage within the FeCN-Nv-2+PAA/vis system (*SI Appendix, Fig. S18*). An observable enhancement in BPA removal occurred as the catalyst dosage was increased from 0.1 g/L to 0.2 g/L. However, a further increment to 0.3 g/L did not lead to a noteworthy acceleration in the reaction rate. Thus, a dosage of 0.2 g/L was determined as the optimal concentration for the subsequent experiments. We conducted a detailed investigation into the effect of the Fe dose on the photocatalytic activity, finding that the FeCN-Nv-2 sample demonstrated optimal performance. However, further increases in Fe content led to a significant decrease in activity in the FeCN-Nv-3 sample, revealing a delicate balance. Synergy was amplified within a specific interval, but problems arose beyond that range, such as carrier recombination. Similarly, PAA concentration was also critical, and the BPA removal kinetics increased rapidly when the PAA dose was increased from 0.01 to 1.00 mM (*SI Appendix, Fig. S16 and Table S12*). However, a further increase to 2.00 mM resulted in no apparent change. Thus, 1 mM PAA was selected as a suitable dose. Such a system also exhibited robust adaptability across a broad pH spectrum (3.0 to 9.0) (*SI Appendix, Fig. S17 and Table S13*), further confirming its versatility and efficiency in contaminant degradation. Furthermore, the concentration of BPA was found to significantly impact the reaction dynamics, as higher concentrations triggered competitive reactions (*SI Appendix, Fig. S20*). We identified the optimal concentration range for all photocatalysts within 0.2 to 0.3 g/L by applying a six-flux model. This value closely matched well the optical thickness range (1.8 to 4.4) of the reactor (*SI Appendix, Fig. S19 and Tables S14 and S15*), aligning with the above experimental results. This critical finding offers the prospect of achieving exact control and fine-tuning catalytic processes, ensuring peak performance and efficacy in real-world applications.

The mineralization in the FeCN-Nv-2+PAA/vis system could attain 63.1% (*SI Appendix, Fig. S21*) and exhibited reduced toxicity for intermediates (Fig. 4D), signifying parallel degradation and mineralization in this system. Remarkably, the stability of the Nv-rich Fe SACs was evidenced by a consistent removal of BPA exceeding 80% over 20 cycles (Fig. 4E). However, this removal efficiency was highly dependent on visible light. In its absence, the BPA removal rate decreased markedly with each successive cycle, plummeting to nearly below 10% after just the 14th cycle (Fig. 4F). These findings indicate the efficiency and stability of the catalyst's reliance on visible light and highlight the importance of incorporating a photoswitching system endowed with self-cleaning attributes for optimal catalytic performance. The FeCN-Nv-2+PAA/vis system exhibited remarkable prowess in degrading BPA and other tenacious pollutants, attesting to its broad-spectrum efficiency (*SI Appendix, Fig. S22*). In fact, the photocatalytic performance of this system surpassed previously reported results (Fig. 4H), suggesting its superiority in handling complex environmental challenges. To extend the inquiry and explore further applications, we tested the system's compatibility with widely used oxidants in AOPs, and peroxymonosulfate (PMS), periodate (PI), and peroxydisulfate (PDS) were selected. Impressively, FeCN-Nv-2 displayed potential

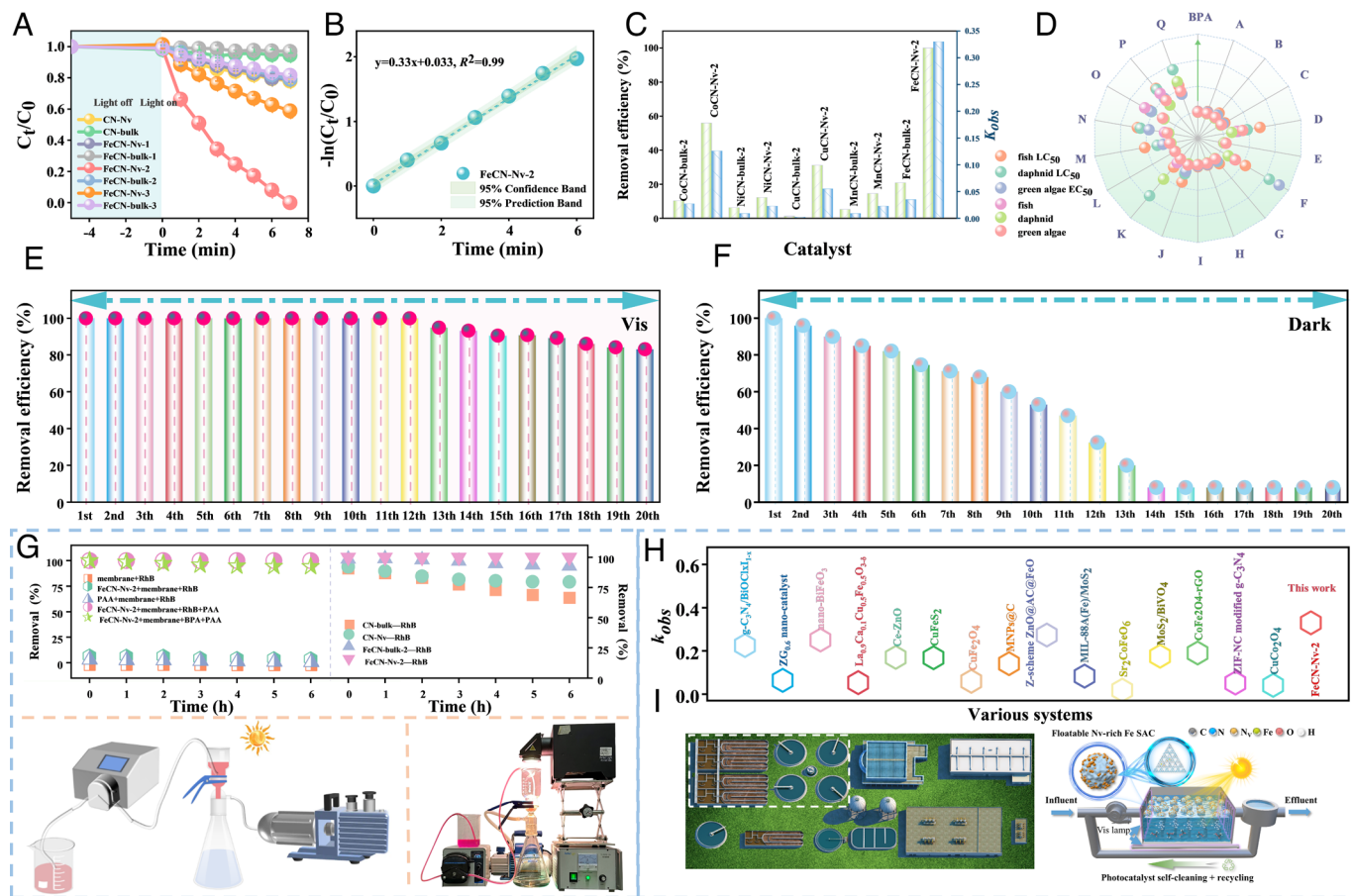


Fig. 4. Efficient and highly stable photo-Fenton-like decontamination. (A and B) Photocatalytic degradation of BPA (A) and the corresponding reaction rate constants in FeCN-Nv-2+PAA reaction systems (B) (Conditions: [Catal.]₀ = 0.2 g/L, [PAA]₀ = 1 mM, [pH]₀ = 6.35, [BPA]₀ = 0.022 mM, [Temp.] = 20 ± 2 °C, the error bars represent the SDs from duplicate tests). (C) Degradation of BPA under visible light irradiation using M-CN-Nv-2 and M-CN-bulk-2 (M = Fe, Co, Mn, Ni, Cu) as photocatalysts. (D) The toxicity evaluation of different intermediates. (E and F) Evaluation of BPA removal using FeCN-Nv-2+PAA system with 20 cycles under light (E) and without light (F). (G) Continuous flow experiment. (H) Comparison of k_{app} values of different systems. (I) A conceptual diagram of a potential WWTP application based on Nv-rich iron Fe SAC-based photo-Fenton-like technology.

in asymmetrical oxidant activation as well (SI Appendix, Fig. S23 and Table S16), highlighting the versatile and promising nature of this innovative catalytic platform.

Nonetheless, it is difficult for the powder catalysts to be recycled in practical applications. To address this issue, we proposed two practicable strategies that could serve as valuable references. First, we anchored Nv-rich Fe SACs onto cost-effective membrane supports, specifically polyvinylidene fluoride (PVDF) membranes. Following vacuum fixation, the empirical results revealed both efficacy and resilience in continuous operations (Fig. 4G and SI Appendix, Figs. S24 and S25). These SACs maintained robust activity throughout extended reaction periods. Second, Nv-rich Fe SACs were incorporated into commercial polystyrene spheres, forming a floatable catalyst (Fig. 4I and SI Appendix, Fig. S49). Importantly, the photocatalyst demonstrated a self-cleaning capability, accomplished through direct light exposure following each cycle. This innovative feature negated the necessity of conventional carbon material regeneration, thus simplifying the process.

Ascertainment of Active Species Involved in the Decontamination.

The observed catalytic efficiency in the FeCN-Nv-2 system led us to explore underlying mechanisms, initially ruling out the potential influence of H₂O₂ in PAA. As shown in SI Appendix, Table S16, the H₂O₂ photoactivation with single FeCN-Nv-2 only removed 11.9% of BPA, affirming that H₂O₂'s contribution was

minimal. Quenching experiments were performed to discern the key reactive species involved (Fig. 5A and SI Appendix, Fig. S27 and Table S11). Dosing TEMPOL ([•]O₂[−] quencher) and L-histidine (¹O₂ quencher) separately into the FeCN-Nv+PAA/vis resulted in a noticeable reduction in activity, highlighting the pivotal roles of [•]O₂[−] and ¹O₂. The introduction of 2, 4-HD, a known quencher of organic radicals (R-O[•]) such as CH₃COO[•] and CH₃COOO[•] (42, 43), did not noticeably affect BPA removal. This observation challenges the previous findings by suggesting an absence of R-O[•]. Crucially, this insight inferred that PAA mainly functioned as an exceptional electron acceptor when integrated into the system. This facilitated the generation of other reactive species, shedding light on the catalytic mechanism. Moreover, adding TBA, EDTA-2Na, and PMSO led to a slight decline in efficiency, ruling out the roles of [•]OH, h⁺, and high-valent iron species. Additionally, the minimal conversion of PMSO to PMSO₂ (less than 1.5%, SI Appendix, Fig. S29) further confirmed that no relevant iron species were generated during the process.

As previously reported, stable 2, 2, 6, 6-tetramethyl-1-piperidinyloxy (TEMPO) could react with R-O[•] to form an organic radical-TEMPO complex. However, in our experiments, no peak indicative of TEMPO [*m/z* = 140.14328] was detected (SI Appendix, Fig. S30). Moreover, the introduction of 0.1 mM TEMPO into the FeCN-Nv-2+PAA/vis system revealed no signal around *m/z* ~ 217, suggesting the absence of CH₃COO[•]-TEMPO (SI Appendix, Fig. S31). Likewise, CH₃[•]-TEMPO signals were also

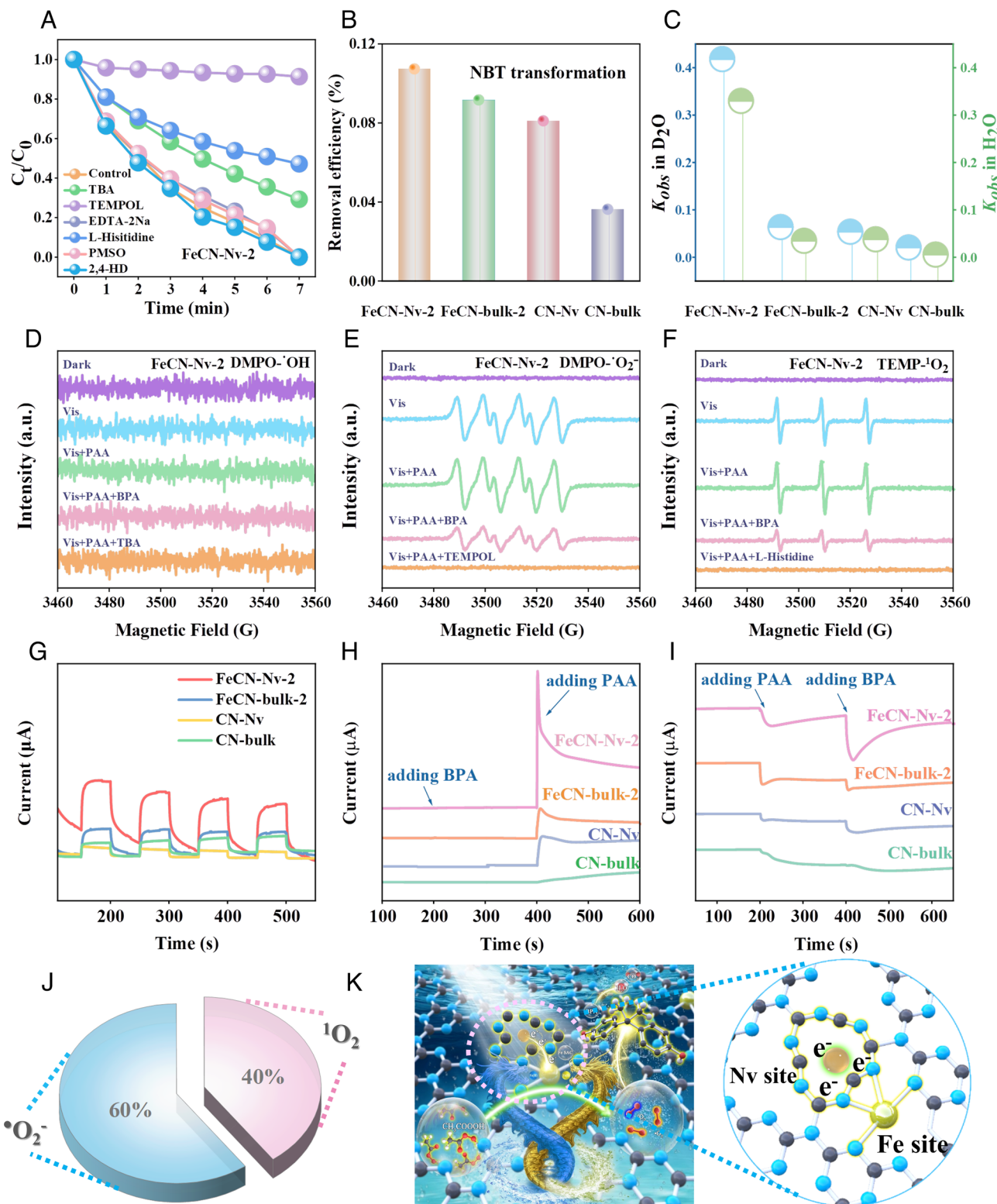


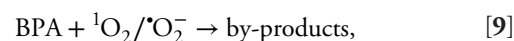
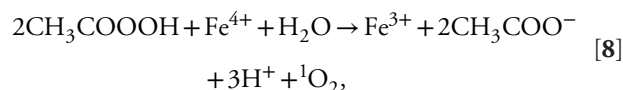
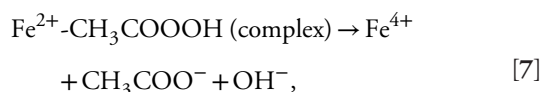
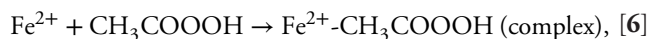
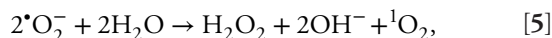
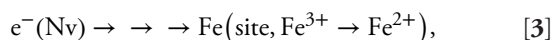
Fig. 5. Identification of the key reactive species. (A) Degradation of BPA in the presence of various scavengers in the FeCN-Nv-2+PAA/vis system. (B) NBT removal using different catalytic systems. (C) Degradation of BPA by different catalysts using D₂O. (D and E) DMPO spin-trapping ESR spectra for (D) DMPO·OH and (E) DMPO·O₂⁻ in the FeCN-Nv-2+PAA/vis system. (F) TEMP-based ESR spectra for ¹O₂. (G–I) i-t curves for different systems. (J) Proportion of active species. (K) Enhanced electron density of Fe SAC by neighboring Nv and a possible photocatalytic mechanism for visible light-assisted PAA activation using FeCN-Nv-2.

missing, possibly due to the rapid conversion of CH₃COO[•] to CH₃[•] (SI Appendix, Fig. S32). Additionally, no signal was detected for the CH₃COOO[•]-TEMPO complex. These findings indicate the absence of R-O[•] in the constructed FeCN-Nv-2+PAA/vis system.

An ESR measurement was then conducted to corroborate the quenching results. No signals attributable to DMPO·OH (44) (Fig. 5D and SI Appendix, Fig. S33A) and TEMPO-h[•] (45) (SI Appendix, Fig. S34 A and B) were found, verifying their comparatively minor contributions. Under visible light irradiation,

the signals related to DMPO- $\cdot\text{O}_2^-$ emerged, and these were notably amplified in the presence of PAA (Fig. 5E). Moreover, dosing BPA and TEMPOL as $\cdot\text{O}_2^-$ scavengers drastically reduced the DMPO- $\cdot\text{O}_2^-$ signals, thus affirming that the $\cdot\text{O}_2^-$ radical was a primary contributor. This observation was further supported by the conversion of nitroblue tetrazolium [NBT (46)] (Fig. 5B and *SI Appendix*, Fig. S35), which underscored the pivotal role of the $\cdot\text{O}_2^-$. Likewise, detecting TEMP- $^1\text{O}_2$ signals validated the presence and significance of $^1\text{O}_2$ (Fig. 5F). *SI Appendix*, Fig. S33 B and C show that DMPO- $\cdot\text{O}_2^-$ and TEMP- $^1\text{O}_2$ signals were markedly intensified with PAA using FeCN-Nv-2, suggesting that Nv might serve as an additional electron donor. To gain further insight into the photocatalytic process, we conducted correlation experiments using distinct solvents, specifically comparing H_2O with D_2O (Deuterium oxide). As displayed in Fig. 5C (*SI Appendix*, Table S18), the photoactivity showed a modest increase when the solvent was switched from H_2O to D_2O , suggesting the involvement of $^1\text{O}_2$. This observation supports the hypothesis that $^1\text{O}_2$ was pivotal in the process. Quantitative normalization indicated that $\cdot\text{O}_2^-$ constituted 60% of the total reactive species, with $^1\text{O}_2$ comprising 40% (Fig. 5J and *SI Appendix*, Text S16).

Based on these experimental results and simulations, we proposed a synergistic enhancement mechanism. When subjected to visible light exposure, electrons in the VB of FeCN-Nv-2 could be excited to transfer to its CB, leaving holes in the VB and accumulating electrons in the CB (Eq. 2). Notably, the presence of Nv within the CN-bulk acted as an effective electron trap. Upon exposure to visible light, it adeptly captured photoexcited electrons, guiding them to the adjacent single-atom Fe sites, enhancing reductive capabilities, and optimizing PAA activation (Eq. 3). Meanwhile, these photogenerated electrons could migrate to the catalyst's surface, interacting with dissolved oxygen to produce $\cdot\text{O}_2^-$ (Eq. 4, Fig. 5G). Further reactions involving $\cdot\text{O}_2^-$ and H_2O molecules amplified the generation of $^1\text{O}_2$ (Eq. 5). Incorporating the oxidant PAA into this reaction system served as an electron sink, substantially improving the spatial separation of charge carriers and expanding the pathways for molecular oxygen activation (Fig. 5H and I). The intrinsic iron cycling process catalyzed PAA activation, promoting $^1\text{O}_2$ production (Eqs. 6–8) and enhancing the system's overall performance. Cumulatively, the FeCN-Nv-2+PAA/vis system expedited decontamination by producing a high yield of $\cdot\text{O}_2^-$ and $^1\text{O}_2$ (Eqs. 9 and 10), thus maximizing the electron reduction pathway on CB and enabling oxygen activation and PAA utilization. A detailed representation of the proposed photocatalytic mechanism is provided in Fig. 5K and *SI Appendix*, Fig. S36.



Intrinsic Catalytic Reaction Mechanism of Photo-Fenton-Like Catalysis. Density functional theory (DFT) calculations were used to further explore the enhancement mechanism at the molecular level, further shedding light on the underlying processes. Initially, a comprehensive analysis was performed on the chemical structure of BPA, including its geometric configuration and electronic properties (Fig. 6A–C and *SI Appendix*, Table S20). Additionally, the condensed Fukui index was calculated and visually represented (Fig. 6D–F), offering a critical insight into the potentially reactive sites within the molecule. These findings enriched our understanding of the underlying mechanisms governing pollutant degradation (*SI Appendix*, Fig. S37).

Then, Fig. 6G–I depicts the two-dimensional electron density maps of CN-bulk, CN-Nv, and FeCN-Nv-2, respectively. The electron-rich and electron-deficient regions are represented by red and blue colors, respectively. The CN-bulk configuration exhibited a highly symmetrical electron distribution, indicating uniform electron density. However, with the introduction of the Nv site, a noticeable shift in electron redistribution occurred, revealing an intriguing alteration in electronic properties. Following the strategic insertion of a single Fe atom site, a substantial electron linkage channel emerged between Fe and the neighboring C atom adjacent to Nv. This innovative structural modification suggested a clear pathway for electron transfer, potentially unlocking broad avenues for enhanced catalytic activity. Moreover, the Gibbs energy calculations, as shown in Fig. 6J, shed light on the energetics of the interaction between PAA and the catalytic system containing FeCN-Nv-2. Upon introduction, the two compounds initially combined to form a critical intermediate complex denoted as “PAA-FeCN-Nv-2.” This process was associated with a Gibbs energy change of -0.52 eV, indicating that this complex's formation was energetically favorable. Such a result supported the observed experimental outcomes and suggested a spontaneous reaction under the provided conditions, offering significant implications for the underlying photocatalytic mechanism.

To shed light on the intricate role of Nv and Fe in molecular activation, the adsorption and activation processes of various molecules, including oxidant (PAA), pollutant (BPA), and oxygen (O_2), were examined (*SI Appendix*, Tables S21 and S22 and Fig. S55). Fig. 6Q unveils the adsorption energy of PAA on CN-bulk, CN-Nv, and FeCN-Nv-2, along with variations in the O–O bond length. The CN-bulk's positive adsorption energy and the O–O bond length, comparable to standard PAA, suggested a challenge in molecular activation due to difficulty in adsorption. Conversely, the introduction of Nv prompted a reduction in adsorption energy and an elongation of the O–O bond, albeit not reaching a significant activation threshold. Strikingly, the construction of FeCN-Nv-2 gave rise to a more negative adsorption energy of -2.88 eV, coupled with a further extension of the O–O bond, reflecting a substantial electron gain from FeCN-Nv-2 (Fig. 6K–M). This robust interaction was mirrored in the adsorption energies of O_2 (-1.35 eV) and BPA (-0.57 eV), as well as in the highest electron transfer toward BPA (Fig. 6N–S and

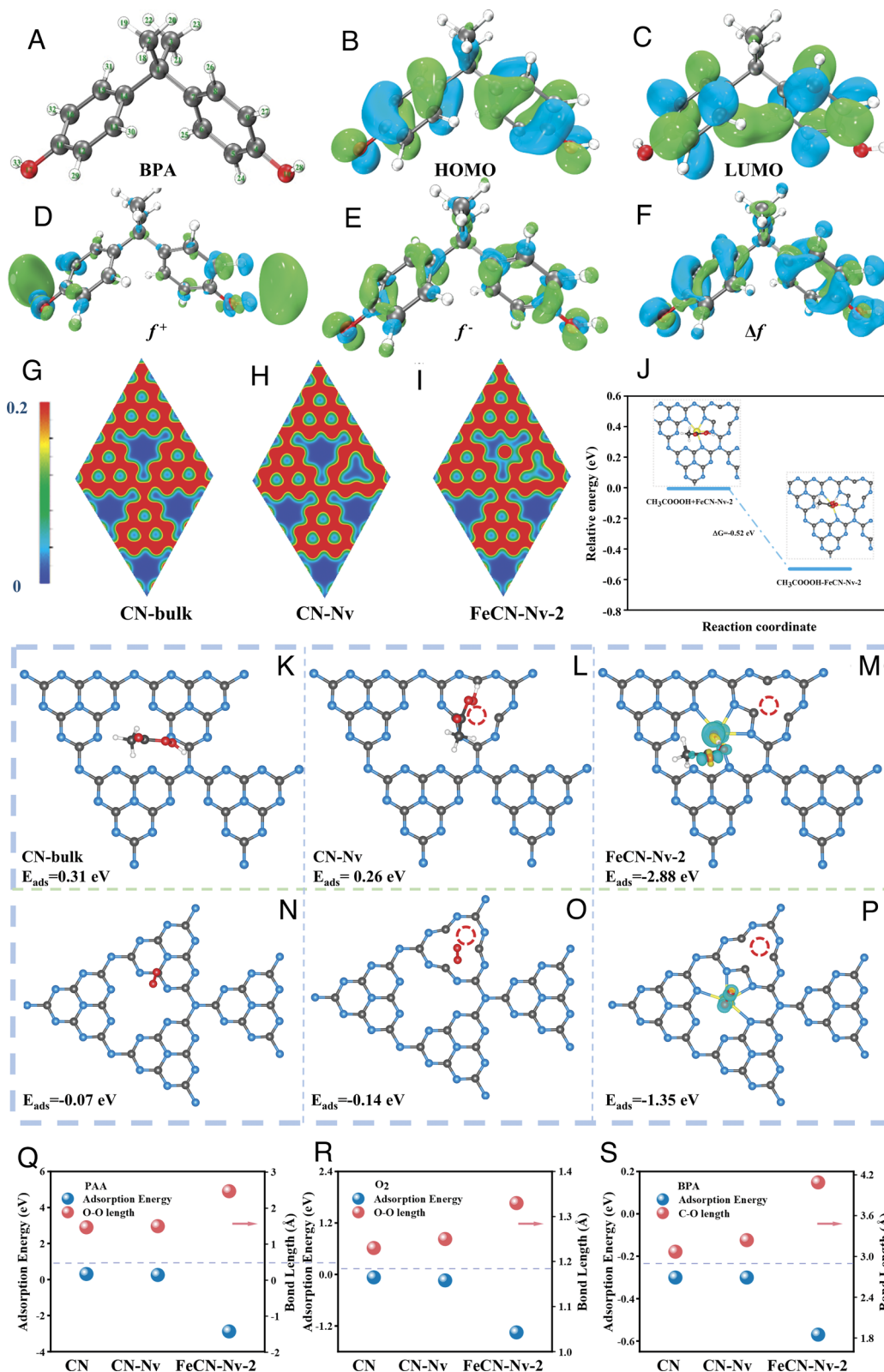


Fig. 6. DFT calculations of the catalytic mechanisms for boosted electron transfer and contamination detoxication. (A–C) The chemical structure of BPA (A), HOMO (B), and LUMO (C) of BPA molecule. (D–F) Condensed Fukui index f^+ (D), f^- (E), and Δf (F) of BPA. (G–I) Two-dimensional charge density: (G) CN-bulk, (H) CN-Nv, and (I) FeCN-Nv-2, respectively. (J) The free energy of PAA adsorption. (K–P) The charge density difference of different catalysts for PAA (K–M) and O₂ (N–P) separately. (Q–S) show the adsorption energies and bond lengths of different catalysts concerning the reaction molecules.

SI Appendix, Fig. S55 A–C), facilitating in situ BPA adsorption enhancement and degradation. These findings substantiate the notion of synergistic activation through dual sites of Fe and Nv,

positioning this design as a potent strategy for pollutant degradation and contributing to a deep understanding of heterogeneous catalysis.

Conclusions

In this work, we successfully tailored Nv-rich Fe SACs by employing a series of computational predictions and evaluating optical properties using a six-flux model, specifically focusing on the apparent optical thickness (τ_{app}). The resultant optimized catalyst was effectively employed to activate PAA under visible light, facilitating the rapid removal of organic pollutants. Central to this design was a unique system that maximized electron utilization, particularly along the PAA reduction pathway on the catalyst's CB, leading to the generation of multiple active species. A key mechanism involved the Nv near single-atom Fe sites continuously providing electrons, thus favoring PAA activation. Notably, this system accomplished complete BPA removal and 63.1% mineralization, echoing the attributes of a photo-Fenton-like reaction. Demonstrating impressive resilience in various environmental conditions, this constructed system effectively degraded multiple pollutants, validating its applicability in real-world scenarios. An intriguing feature was its ability to achieve self-cleaning in powder form and fixed state over 20 continuous cycles. A noteworthy finding was that residual PAA might enhance the acetic acid concentration in water, potentially contributing as a carbon source for subsequent biological treatments (*SI Appendix, Figs. S61–S63 and Text S19 and S20*). This work paves the way for innovative strategies in organic wastewater treatment and other high-value-added applications, emphasizing a nuanced modulation of local electron density at catalytic sites and the optimal configuration of reaction systems.

1. Z. Wang *et al.*, Cobalt single atoms anchored on oxygen-doped tubular carbon nitride for efficient peroxymonosulfate activation: Simultaneous coordination structure and morphology modulation. *Angew Chem. Int. Ed.* **61**, e202202338 (2022).
2. Z. Y. Guo *et al.*, Mn-O covalency governs the intrinsic activity of Co-Mn spinel oxides for boosted peroxymonosulfate activation. *Angew Chem. Int. Ed.* **60**, 274–280 (2021).
3. Y. J. Zhang *et al.*, Simultaneous nanocatalytic surface activation of pollutants and oxidants for highly efficient water decontamination. *Nat. Commun.* **13**, 3005 (2022).
4. R. Li *et al.*, Peracetic acid-ruthenium(III) oxidation process for the degradation of micropollutants in water. *Environ. Sci. Technol.* **55**, 9150–9160 (2021).
5. B. Liu *et al.*, Novel nonradical oxidation of sulfonamide antibiotics with Co(II)-doped g-C₃N₄-activated peracetic acid: Role of high-valent cobalt-oxo species. *Environ. Sci. Technol.* **55**, 12640–12651 (2021).
6. B. Wang *et al.*, A site distance effect induced by reactant molecule matchup in single-atom catalysts for Fenton-like reactions. *Angew. Chem. Int. Ed.* **61**, e202207268 (2022).
7. J. Kim *et al.*, Advanced oxidation process with peracetic acid and Fe(II) for contaminant degradation. *Environ. Sci. Technol.* **53**, 13312–13322 (2019).
8. Z. Wang *et al.*, Application of cobalt/peracetic acid to degrade sulfamethoxazole at neutral condition: Efficiency and mechanisms. *Environ. Sci. Technol.* **54**, 464–475 (2020).
9. R. A. Kenley, T. G. Traylor, Radical-induced decomposition of peracetic acid. *J. Am. Chem. Soc.* **97**, 4700–4706 (1975).
10. F. Chen *et al.*, Single-atom iron anchored tubular g-C₃N₄ catalysts for ultrafast Fenton-Like reaction: Roles of high-valent iron-oxo species and organic radicals. *Adv. Mater.* **34**, 2202891 (2022).
11. M. Cai, P. Sun, L. Zhang, C. H. Huang, UV/peracetic acid for degradation of pharmaceuticals and reactive species evaluation. *Environ. Sci. Technol.* **51**, 14217–14224 (2017).
12. X. Mi *et al.*, Almost 100% peroxymonosulfate conversion to singlet oxygen on single-atom CoN₂+2 sites. *Angew. Chem. Int. Ed.* **133**, 4638–4643 (2020).
13. L. S. Zhang *et al.*, Carbon nitride supported high-loading Fe single-Atom catalyst for activation of Peroxymonosulfate to generate ¹O₂ with 100% selectivity. *Angew. Chem. Int. Ed.* **60**, 21751–21755 (2021).
14. E. V. Rokhina, K. Makarova, E. A. Golovina, H. Van As, J. Virkutyte, Free radical reaction pathway, thermochemistry of peracetic acid homolysis, and its application for phenol degradation: Spectroscopic study and quantum chemistry calculations. *Environ. Sci. Technol.* **44**, 6815–6821 (2010).
15. K. B. Newhart *et al.*, Prediction of peracetic acid disinfection performance for secondary municipal wastewater treatment using artificial neural networks. *ACS ES&T Water* **1**, 328–338 (2020).
16. B. Liu *et al.*, Insights into the oxidation of organic contaminants by Co(II) activated peracetic acid: The overlooked role of high-valent cobalt-oxo species. *Water Res.* **201**, 117313 (2021).
17. J. W. Wang *et al.*, Applying a novel advanced oxidation process of activated peracetic acid by CoFe₂O₄ to efficiently degrade sulfamethoxazole. *Appl. Catal. B-Environ.* **280**, 119422 (2021).
18. S. R. Yang *et al.*, Efficient activation of PAA by FeS for fast removal of pharmaceuticals: The dual role of sulfur species in regulating the reactive oxidized-species. *Water Res.* **217**, 118402 (2022).
19. Y. Xiong *et al.*, Single-atom Fe catalysts for Fenton-like reactions: Roles of different N species. *Adv. Mater.* **34**, e2110653 (2022).
20. C. Cao *et al.*, Fe₃O₄/Ag/Bi₂MoO₆ photoactivatable nanzyme for self-replenishing and sustainable cascaded nanocatalytic cancer therapy. *Adv. Mater.* **33**, e2106996 (2021).

Materials and Methods

Chemicals, Synthesis, Characterizations, Measurements, and Theoretical Calculation. Detailed information is provided in *SI Appendix, Text S1–S9*.

Catalytic Performance Evaluations. All degradation experiments were conducted using a 50 mL suspension containing 0.2 g/L of the catalyst, 1 mM PAA, and 0.022 mM BPA, with an aqueous temperature maintained at 20 ± 2 °C via circulating water. Before initiating the photocatalytic reaction, the mixture of catalyst and BPA was stirred for 10 min to achieve adsorption-desorption equilibrium. A 300 W Xenon lamp (PLS-SXE 300D/300DUV, Beijing Perfectlight Co.) with a 420-nm cutoff filter served as the light source. Upon switching on the light, PAA (1 mM) was promptly added to initiate the reaction. Samples of 1 mL were taken from the solution and immediately mixed with prepared ascorbic acid (0.1 mL, 400 mM) to halt the reaction, followed by centrifugation at 14,000 rpm for 10 min for solution collection. Residual concentration and intermediates of BPA were then detected using high-performance liquid chromatography coupled with mass spectrometry (HPLC-MS, Agilent Co.).

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

ACKNOWLEDGMENTS. We thank the National Natural Science Foundation of China (52270149 and 51908528) and Fundamental Research Funds for the Central Universities grant (2021CDJQY-014) for supporting this work.

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21. C. Liu *et al.*, Ternary MOF-on-MOF heterostructures with controllable architectural and compositional complexity via multiple selective assembly. *Nat. Commun.* **11**, 4971 (2020).
22. C. Gao *et al.*, Heterogeneous single-atom photocatalysts: Fundamentals and applications. *Chem. Rev.* **120**, 12175–12216 (2020).
23. J. W. Fu, J. G. Yu, C. J. Jiang, B. Cheng, g-C₃N₄-based heterostructured photocatalysts. *Adv. Energy Mater.* **8**, 1701503 (2018).
24. F. Chen *et al.*, Efficient decontamination of organic pollutants under high salinity conditions by a nonradical peroxymonosulfate activation system. *Water Res.* **191**, 116799 (2021).
25. L. L. Liu *et al.*, Edge electronic vacancy on ultrathin carbon nitride nanosheets anchoring O₂ to boost H₂O₂ photoproduction. *Appl. Catal. B: Environ.* **302**, 120845 (2022).
26. J. Xiao, Y. Xie, J. Rabeah, A. Bruckner, H. Cao, Visible-light photocatalytic ozonation using graphitic C₃N₄ catalysts: A hydroxyl radical manufacturer for wastewater treatment. *Acc. Chem. Res.* **53**, 1024–1033 (2020).
27. J. F. Yu *et al.*, Metal-free carbon materials for persulfate-based advanced oxidation process: Microstructure, property and tailoring. *Prog. Mater. Sci.* **111**, 100654 (2020).
28. W. J. Ong, L. L. Tan, Y. H. Ng, S. T. Yong, S. P. Chai, Graphitic carbon nitride (g-C₃N₄)-based photocatalysts for artificial photosynthesis and environmental remediation: Are we a step closer to achieving sustainability? *Chem. Rev.* **116**, 7159–7329 (2016).
29. W. Liu *et al.*, Visible-light-driven photocatalytic degradation of diclofenac by carbon quantum dots modified porous g-C₃N₄: Mechanisms, degradation pathway and DFT calculation. *Water Res.* **151**, 8–19 (2019).
30. H. Li, C. Shan, B. Pan, Fe(III)-doped g-C₃N₄ mediated peroxymonosulfate activation for selective degradation of phenolic compounds via high-valent iron-oxo species. *Environ. Sci. Technol.* **52**, 2197–2205 (2018).
31. L. B. Jiang *et al.*, Doping of graphitic carbon nitride for photocatalysis: A review. *Appl. Catal. B-Environ.* **217**, 388–406 (2017).
32. Z. H. Lin *et al.*, Atomic Co decorated free-standing graphene electrode assembly for efficient hydrogen peroxide production in acid. *Energy Environ. Sci.* **15**, 1172–1182 (2022).
33. Z. Y. Teng *et al.*, Atomically dispersed antimony on carbon nitride for the artificial photosynthesis of hydrogen peroxide. *Nat. Catal.* **4**, 374–384 (2021).
34. C. Xia *et al.*, General synthesis of single-atom catalysts with high metal loading using graphene quantum dots. *Nat. Chem.* **13**, 887–894 (2021).
35. J. Jiang *et al.*, Nitrogen vacancy-modulated peroxymonosulfate nonradical activation for organic contaminant removal via high-valent cobalt-oxo species. *Environ. Sci. Technol.* **56**, 5611–5619 (2022).
36. Y. Yang *et al.*, Molecular engineering of polymeric carbon nitride for highly efficient photocatalytic oxytetracycline degradation and H₂O₂ production. *Appl. Catal. B-Environ.* **272**, 118970 (2020).
37. I. F. Teixeira, E. C. M. Barbosa, S. C. E. Tsang, P. H. C. Camargo, Carbon nitrides and metal nanoparticles: From controlled synthesis to design principles for improved photocatalysis. *Chem. Soc. Rev.* **47**, 7783–7817 (2018).
38. T. Wang, C. Y. Nie, Z. M. Ao, S. B. Wang, T. C. An, Recent progress in g-C₃N₄ quantum dots: Synthesis, properties and applications in photocatalytic degradation of organic pollutants. *J. Mater. Chem. A* **8**, 485–502 (2020).

39. W. Wang, J. C. Yu, Z. Shen, D. K. Chan, T. Gu, g-C₃N₄ quantum dots: Direct synthesis, upconversion properties and photocatalytic application. *Chem. Commun.* **50**, 10148–10150 (2014).
40. Y. Yang *et al.*, Boron nitride quantum dots decorated ultrathin porous g-C₃N₄: Intensified exciton dissociation and charge transfer for promoting visible-light-driven molecular oxygen activation. *Appl. Catal. B-Environ.* **245**, 87–99 (2019).
41. Z. Z. Ai *et al.*, Rational modulation of p-n homojunction in P-doped g-C₃N₄ decorated with Ti₃C₂ for photocatalytic overall water splitting. *Appl. Catal. B-Environ.* **259**, 118077 (2019).
42. J. Kim, J. Wang, D. C. Ashley, V. K. Sharma, C. H. Huang, Enhanced degradation of micropollutants in a peracetic acid-Fe(III) system with picolinic Acid. *Environ. Sci. Technol.* **56**, 4437–4446 (2022).
43. T. Liu *et al.*, Unexpected role of nitrite in promoting transformation of sulfonamide antibiotics by peracetic acid: Reactive nitrogen species contribution and harmful disinfection by-product formation potential. *Environ. Sci. Technol.* **56**, 1300–1309 (2022).
44. C. Ling *et al.*, Atomic-layered Cu^S nanoclusters on FeS₂ with dual catalytic sites for efficient and selective H₂O₂ activation. *Angew. Chem. Int. Ed.* **61**, e202200670 (2022).
45. C. X. Zhao *et al.*, Probing supramolecular assembly and charge carrier dynamics toward enhanced photocatalytic hydrogen evolution in 2D graphitic carbon nitride nanosheets. *Appl. Catal. B: Environ.* **256**, 117867 (2019).
46. F. Chen *et al.*, Catalytic degradation of ciprofloxacin by a visible-light-assisted peroxymonosulfate activation system: Performance and mechanism. *Water Res.* **173**, 115559 (2020).